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310407

Roll No. _____ Total No of Pages: 2

310407

B. Tech. III - Sem. (Main) Exam., February - 2021
Computer Science & Engineering
3CS4-07 Software Engineering
Common for CS/ IT

Time: 3 Hours **Maximum Marks: 120**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions (up to 100 words) 5 × 8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 × 15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

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PART - A

Q.1 Distinguish clearly between verification & validation.

Q.2 Define software process. State the important features of a process.

Q.3 How do you estimate time required for a software development project?

Q.4 List out the importance of cost estimation in software development.

Q.5 What is known as SRS review? How is it conducted?

Q.6 What is data dictionary? How is it used in software engineering?

Q.7 What is meant by information flow continuity?

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Q.8 What is cardinality in data modeling? What does modality in data modeling indicate?

Q.9 What is data modeling? What is a data object?

Q.10 Explain the object modularization with example.

PART - B

Q.1 Describe various prototyping techniques and discuss onanalysis and modeling.

Q.2 Explain the feasibility studies. What are the outcomes? Does it have either implicit or explicit effects on software requirement collection.

Q.3 What are the characteristics of a good design? Describe different types of coupling and cohesion. How design evaluation is performed?

Q.4 What is the impact of requirement changes during development of a software product?

Q.5 Explain UML with all its diagram.

Q.6 Explain the state oriented approaches for representing behavioral specifications of software. https://www.btubikaner.com

Q.7 Discuss Object Oriented Analysis (OOA) and modeling in detail.

PART - C

Q.1 List several software process paradigms. Explain how both waterfall model and prototyping model can be accommodated in the spiral process model.

Q.2 List out the various activities that are encompassed by system design process under Object Oriented Design (OOD) and explain each one briefly.

Q.3 Explain various Cost estimation models and compare.

Q.4 Explain about the various design concepts considered during design. Write short notes on user interface design process.

Q.5 Compare functional and behavioral models. With a suitable diagram explain the elements of the analysis model. Also explain about DFD.

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